

Volume of prisms and cylinders



1 1728 cm^3

2 3 cm

3 9 cm

4 6 m

5

a 48 cm^3

b 32 cm^3

c 217.5 cm^3

d 330 cm^3

e 32 cm^3

f 270 cm^3

g 756 cm^3

h 63 cm^3

i 1800 cm^3

j 187.738 cm^3

k 399 cm^3

6

a No

b No

c No

d Yes

7

a 8 cm

b 768 cm^3

8 59.7 cm^3

9 2309.07 cm

10

a 294.5 cm^3

b 307.9 cm^3

c 70.7 cm^3

d 816.8 cm^3

e 263.9 cm^3

f 1608.5 cm^3

g 1057.9 cm^3

h 755.9 cm^3

i 940.1 cm^3

j 883.6 cm^3

k 491.5 cm^3

11

a 552.92 cm^3

b 2211.68 cm^3

c 15.5 L

12 829.38 cm^3

13

a 131.43 cm^2

b 1445.73 cm^3



Additional questions

14 $r = 2.3 \text{ cm}$

15

a Volume = $400\pi \text{ cm}^3$

b height = 20 cm

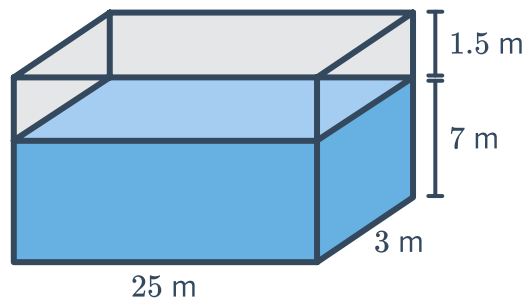
16 8 die

17

a 183.6 cm^3

b 156.06 cm^3

18 Any dimensions where the water occupies 524.6 m^3 of space with an extra 1.5 m clearance at the top of the tank. For example,



19 Either option will fit the volume of all of items. The cost of purchasing $15 \times$ Box 1 is slightly cheaper than the cost of $36 \times$ Box 2 so if you want to save money you can select the option to buy $15 \times$ Box 1.